

Dr. Xiaoyue Liu

Postdoctoral Researcher, Barcelona Institute for Global Health (ISGlobal)

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[ORCID](#) | [Web of Science](#) | [Google Scholar](#)

Interdisciplinary researcher developing climate-informed health risk assessments and early-warning systems across spatial scales and prediction horizons. My research aims to support targeted public-health decision-making and protective action.

POSITIONS

Barcelona Institute for Global Health (ISGlobal)

Postdoctoral Researcher

Barcelona, Spain

2026.05–Present

National University of Singapore, Saw Swee Hock School of Public Health

Research Fellow

Singapore

2024.08–2025.04

Lanzhou University, College of Atmospheric Sciences

Research Assistant

Lanzhou, China

2018.09–2020.06

FUNDINGS

- **Marie Skłodowska-Curie Actions Postdoctoral Fellowship**, European Commission. MSCA Postdoctoral Fellow; Grant Agreement No. 101277999; 2027–2029; EUR 194,074.56.
- **Juan de la Cierva Fellowship**, Spanish State Research Agency (AEI), Ministry of Science, Innovation and Universities; co-funded by the European Social Fund Plus. Juan de la Cierva Fellow; reference JDC2025-057886-I; 2026–2029; EUR 120,000.

EDUCATION

Lanzhou University, College of Atmospheric Sciences

Ph.D. in Climatology

Lanzhou, China

2018.09–2024.06

Doctoral advisor: *Prof. Jianping Huang, Member of the Chinese Academy of Sciences*

Lanzhou University, College of Atmospheric Sciences

B.S. in Atmospheric Sciences

Lanzhou, China

2014.09–2018.06

Academic advisor: *Prof. Jianping Huang*

COLLABORATIVE PROJECTS

- **CATALYSE**, Horizon Europe; project period 2023–2027; consortium budget EUR 10,354,021.
Postdoctoral Researcher since May 2025.
- **EARLY-ADAPT**, European Research Council Consolidator Grant; project period 2021–2026; project budget EUR 2,000,000.
Postdoctoral Researcher, May 2025–Jan. 2026.
- **FORECAST-AIR**, European Research Council Proof of Concept; project period 2024–2025; project budget EUR 150,000.
Postdoctoral Researcher, May–Sep. 2025.

ACADEMIC SERVICE

Guest Editor 2025.02–Present
Special Issue, *Climate, Ecosystem and Human Health: Impacts and Adaptation, Climate*.

Peer Reviewer 2020.05–Present
Nature Health, Environmental Research Letters, Scientific Data, Environmental Research, BMC Public Health, Science China Earth Sciences, Journal of Meteorological Research, and others.

HONORS AND AWARDS

First Prize Scholarship for Doctoral Students (Top 10%), Lanzhou University 2023, 2020

National Scholarship for Doctoral Students (Top 1%), Ministry of Education of China 2021

Baosteel Outstanding Student Special Scholarship (25 recipients nationwide), Baosteel Group 2021

Merit Graduate Student (Top 5%), Lanzhou University 2021

First Prize Scholarship for Graduate Students (Top 10%), Lanzhou University 2019, 2018

Meritorious Winner, Mathematical Contest in Modeling (Top 7%) 2016

PRESENTATIONS

Oral Presentations

From Weather Predictions to Health Protection: A Fine-Scale Heat-Health Warning System for Catalonia, *Horizon Europe CATALYSE Project Consortium Meeting*, Faculty of Medicine, University of Porto, Porto, Portugal, Feb., 2026

Fine-Scale Epidemiological Assessment of Heat-Related Mortality in Catalonia, *CATALYSE Online Webinar on Heat and Health System*, Online, Dec., 2025

Investigating Climate and Environmental Changes and Tropical Diseases, *Research Institute for Tropical Medicine, Department of Parasitology Annual Stakeholders' and Feedback Meeting*, Puerto Princesa City, Philippines, Dec. 2024

Harmonizing Climate Data for Enhanced Vector-Borne Disease Management, *Infectious Disease Modelling Conference*, Bangkok, Thailand, Nov., 2024

O₂ Footprint and Dryland Sustainability, *The International Workshop on Geography and Sustainability 2023*, Beijing, China, Oct., 2023

‘Urban Respiration’: Insights from O₂ Measurements, *The 4th Atmospheric Oxygen Workshop*, Brunswick, Maine, United States, Aug., 2023

Urban O₂ Measurements in a Semi-arid Downtown, *The 8th Youth Geoscience Forum*, Wuhan, Hubei, China, May, 2023

Poster Presentation

Global Prediction System for the COVID-19 Pandemic, *Climatological, Meteorological and Environmental Factors in the COVID-19 Pandemic (online symposium)*, WMO, Aug. 2020

PUBLICATIONS

more than 30 peer-reviewed papers; 1,033 citations; H-index 18 (metrics updated July 2026). Full publication records: [ORCID](#) | [Web of Science](#) | [Google Scholar](#).

First-authored

- 7 **Liu, X.**; Quijal-Zamorano, M.; Satorra, P.; Gallo, E.; Shartova, N.; Achebak, H.; Paniello-Castillo, B.; Janoš, T.; Chen, Z.-Y.; Casellas, E.; de’ Donato, F.; Antó, J. M.; Martínez-Beneito, M. A.; Marí-Dell’Olmo, M.; Moré, J.; Ballester, J. Small-area impact-based early warnings for temperature-related mortality: bringing actionable evidence to the last mile. *Nature Climate Change*, under review.
- 6 **Liu, X.**; Masselot, P.; Gasparrini, A.; Ballester, J. Bayesian Spatial Framework for Multi-location Analysis of Environmental Risks. In preparation.
- 5 **Liu, X.**; Wang, L.; Huang, J.; Wang, Y.; Li, C.; Ding, L.; Lian, X.; Shi, J. Revealing the Covariation of Atmospheric O₂ and Pollutants in an Industrial Metropolis by Explainable Machine Learning. *Environmental Science and Technology Letters*, 2023, 10 (10), 851-858.<https://doi.org/10.1021/acs.estlett.3c00505>.
[Journal Cover](#)
- 4 **Liu, X.**; Huang, J.; Wang, L.; Lian, X.; Li, C.; Ding, L.; Wei, Y.; Chen, S.; Wang, Y.; Li, S.; Shi, J. “Urban Respiration” Revealed by Atmospheric O₂ Measurements in an Industrial Metropolis. *Environmental Science and Technology*, 2023, 57 (6), 2286–2296.<https://doi.org/10.1021/acs.est.2c07583>.
[Journal Cover](#)
- 3 **Liu, X.**; Huang, J.; Li, C.; Zhao, Y.; Wang, D.; Huang, Z.; Yang, K. The Role of Seasonality in the Spread of COVID-19 Pandemic. *Environmental Research*, 2021, 195, 110874.<https://doi.org/10.1016/j.envres.2021.110874>. **ESI Highly Cited Paper (Top 1%)**
- 2 **Liu, X.**; Huang, J.; Huang, J.; Li, C.; Ding, L.; Meng, W. Estimation of Gridded Atmospheric Oxygen Consumption from 1975 to 2018. *Journal of Meteorological Research*, 2020, 34 (3), 646–658.<https://doi.org/10.1007/s13351-020-9133-7>.

- 1 **Liu, X.**; Yu, H.; Sheng, X.; Zhu, C.; Zhao, Q.; Ma, Y.; Ju, S. Mechanism Analysis of a Rare “Thunder Snow” Process in Semi-Arid Area. *Meteor. Mon.*, 2020, 46 (12), 1596–1607. <https://doi.org/10.7519/j.issn.1000-0526.2020.12.007>. (in Chinese)

Co-authored

- 39 Ballester, J.; Ruiz-Cabrejos, J.; Paniello-Castillo, B.; Thompson, A. T.; **Liu, X.**; Chen, Z.-Y.; Janoš, T.; Basagaña, X. Temperature-related mortality in Ukraine during 2003–2022. *Communications Health*, under review.
- 38 Natal, S.; Maia, C.; Ballester, J.; Brattich, E.; Czajkowski, M.; Di Napoli, C.; Di Sabatino, S.; Duane, S.; Gao, C.; Jankin, S.; Leonard, A. F. C.; **Liu, X.**; Lowe, R.; Maia, M.; Palamarchuk, Y.; Preet, R.; Robins, P. E.; Ruggieri, P.; Scoccimarro, E.; Semenza, J. C.; Sofiev, M.; Toftum, J.; Urquiza, D.; Wetterhall, F.; San Martin, D. Climate-informed health early warning systems in Europe: current innovations and future directions. *Environment International*, under review.
- 37 Chen-Xu, J.; Kriit, H. K.; Semenza, J. C.; Klauber-Heiliger, H.; Markandya, A.; Dasandi, N.; Jankin, S.; van Daalen, K. R.; Auclair, O.; Ballester, J.; Bechara, H.; Bussalleu, A.; Caleffi, S.; Callaghan, M. W.; Chen, Z.-Y.; Cirach, M.; Coan, T. G.; Crippa, M.; Dasgupta, S.; de’Donato, F.; de Hoogh, K.; Di Pirro, E.; Fransson, P.; Gatland, D.; Gopal, I.; Guizzardi, D.; Hänninen, R.; Hatfield, C. R. S.; Kiese Wetter, G.; Lindl, F.; **Liu, X.**; Malla, R.; Martinez-Urtaza, J.; Minx, J. C.; Montana, F.; Mueller, N.; Nieuwenhuijsen, M.; Nobile, F.; O’Neill, S.; Palamarchuk, J.; Paniello-Castillo, B.; Pisoni, E.; Rawden, J.; Singh, P.; Sjödin, H.; Sofiev, M.; Springmann, M.; Stucki, L.; Triñanes, J.; van den Bosch, M.; Vermeulen, R.; Vielma, C.; Walawender, M.; Romanello, M.; Antó, J. M.; Nilsson, M.; Rocklöv, J.; Tonne, C. The 2026 Europe Cities Report of the Lancet Countdown on health and climate change: cities at the forefront of climate change action. *The Lancet Public Health*, in press.
- 36 Ruan, W.; Zhao, Q.; **Liu, X.**; Ang, L.; Sun, H. Z.; Seow, W. J. Developing a global integrated environmental quality index for public health research. *Environmental Pollution* 2026, 128701. <https://doi.org/10.1016/j.envpol.2026.128701>.
- 35 Ouyang, Z.; Li, C.; Huang, J.; Wang, Y.; Ren, Y.; Shi, J.; **Liu, X.** Ecological feedbacks modulate lake surface oxygen responses to heatwaves. *Science China Earth Sciences* 2026, 69 (4), 1294–1305. <https://doi.org/10.1007/S11430-025-1840-6>.
- 34 Masse, R. S.; Vythilingam, I.; Fornace, K.; Othman, H.; **Liu, X.**; Jaafar, A. J.; Shahar, M. K.; Rahman, N. A. A.; Azman, A. H. K.; Jeyaprakasam, N. K. Impact of environmental factors on the bionomics of *Anopheles* mosquito vectors of zoonotic malaria: A narrative review. *One Health* 2025, 21, 101141. <https://doi.org/10.1016/J.ONEHLT.2025.101141>.
- 33 Han, D.; Huang, J.; **Liu, X.**; Fu, L.; Ding, L.; Zhang, G.; Li, C.; Yang, F.; Shi, J.; Zhang, B. Silent soils have been awakened by climate change. *Catena* 2025, 252, 108777. <https://doi.org/10.1016/J.CATENA.2025.108777>.
- 32 Fu, L.; Huang, J.; Zhang, G.; Han, D.; Ding, L.; Wei, Y.; **Liu, X.**; Li, C.; Yu, H. Abrupt loss of species richness caused by ecosystem transition. *Science Bulletin* 2025, 70 (9), 1523–1532. <https://doi.org/10.1016/j.scib.2025.03.001>.
- 31 Surendra, H.; Djaafara, B.; Prameswari, H.; Supriyanto, D.; Waluyo, P.; Basuki, S.; Herdiana, H.; Ndoen, E.; Siswanto, S.; Lubis, I.; **Liu, X.**; Mishra, S.; Fornace, K.; Elyazar, I. Mitigating risks of malaria and other vector-borne diseases in the new capital city of Indonesia. *Nature Communications* 2024, 15, 10575. <https://doi.org/10.1038/s41467-024-54891-x>.

- 30 Li, C.; Huang, J.; **Liu, X.**; Ding, L.; He, Y.; Xie, Y. The ocean losing its breath under the heatwaves. *Nature Communications* 2024, 15, 6840. <https://doi.org/10.1038/s41467-024-51323-8>.
- 29 Wang, L.; Zhao, Y.; **Liu, X.**; Shi, J. Enhancement of Atmospheric Oxidation Capacity Induced Co-pollution of the O₃ and PM_{2.5} in Lanzhou, Northwest China. *Environmental Pollution* 2023, 122951. <https://doi.org/10.1016/j.envpol.2023.122951>.
- 28 Huang, J.; Zhang, L.; Chen, B.; **Liu, X.**; Yan, W.; Zhao, Y.; Chen, S.; Lian, X.; Liu, C.; Wang, R.; Gao, S.; Wang, D. Development of the Second Version of the Global Prediction System for Epidemiological Pandemic. *Fundamental Research* 2023, S2667325823002303. <https://doi.org/10.1016/j.fmre.2023.02.030>.
- 27 Zhang, G.; Huang, J.; **Liu, X.**; Guan, X.; Wei, Y.; Ding, L.; Han, D. The Evolution of Ecological Security and Its Drivers in the Yellow River Basin. *Environ. Sci. Pollut. Res.* 2023, 30 (16), 47501–47515. <https://doi.org/10.1007/s11356-023-25667-5>.
- 26 Wang, L.; Zhao, Y.; Shi, J.; Ma, J.; **Liu, X.**; Han, D.; Gao, H.; Huang, T. Predicting Ozone Formation in Petrochemical Industrialized Lanzhou City by Interpretable Ensemble Machine Learning. *Environ. Pollut.* 2023, 318, 120798. <https://doi.org/10.1016/j.envpol.2022.120798>.
- 25 Liu, C.; Huang, J.; Chen, S.; Wang, D.; Zhang, L.; **Liu, X.**; Lian, X. The Impact of Crowd Gatherings on the Spread of COVID-19. *Environ. Res.* 2022, 213, 113604. <https://doi.org/10.1016/j.envres.2022.113604>.
- 24 Ding, L.; Huang, J.; Li, C.; Han, D.; **Liu, X.**; Li, H.; Bai, Y.; Huang, J. Variations in Terrestrial Oxygen Sources under Climate Change. *Sci. China Earth Sci.* 2022, 65 (9), 1810–1823. <https://doi.org/10.1007/s11430-021-9956-5>.
- 23 Li, C.; Huang, J.; Ding, L.; Ren, Y.; An, L.; **Liu, X.**; Huang, J. The Variability of Air-Sea O₂ Flux in CMIP6: Implications for Estimating Terrestrial and Oceanic Carbon Sinks. *Adv. Atmos. Sci.* 2022, 39 (8), 1271–1284. <https://doi.org/10.1007/s00376-021-1273-x>.
- 22 Huang, J.; Lian, X.; Zhao, Y.; Wang, D.; Chen, S.; Zhang, L.; **Liu, X.**; Gao, J.; Liu, C. Water Transmission Increases the Intensity of COVID-19 Outbreaks. *Front. Public Health* 2022, 10, 808523. <https://doi.org/10.3389/fpubh.2022.808523>.
- 21 Ren, Y.; Yu, H.; Liu, C.; He, Y.; Huang, J.; Zhang, L.; Hu, H.; Zhang, Q.; Chen, S.; **Liu, X.**; Zhang, M.; Wei, Y.; Yan, Y.; Fan, W.; Zhou, J. Attribution of Dry and Wet Climatic Changes over Central Asia. *J. Climate* 2022, 35 (5), 1399–1421. <https://doi.org/10.1175/JCLI-D-21-0329.1>.
- 20 Li, S.; Sato, T.; Nakamura, T.; **Liu, X.**; Guo, W. Controlling Factors of Historical Variation of Winter Tibetan Plateau Snow Cover Revealed by Large-Ensemble Experiments. *JGR Atmospheres* 2021, 126 (21). <https://doi.org/10.1029/2021JD035127>.
- 19 Han, D.; Huang, J.; Ding, L.; **Liu, X.**; Li, C.; Yang, F. Oxygen Footprint: An Indicator of the Anthropogenic Ecosystem Changes. *CATENA* 2021, 206, 105501. <https://doi.org/10.1016/j.catena.2021.105501>.
- 18 Wei, Y.; Yu, H.; Huang, J.; **Liu, X.**; Zhou, J. Improving China's Summer Precipitation Prediction in 2020 by Observational Constrained Bias Correction. *Theor. Appl. Climatol.* 2021, 145 (3–4), 1317–1331. <https://doi.org/10.1007/s00704-021-03693-y>.

- 17 Liu, C.; Huang, J.; Ji, F.; Zhang, L.; **Liu, X.**; Wei, Y.; Lian, X. Improvement of the Global Prediction System of the COVID-19 Pandemic Based on the Ensemble Empirical Mode Decomposition (EEMD) and Autoregressive Moving Average (ARMA) Model in a Hybrid Approach. *Atmospheric and Oceanic Science Letters* 2021, 14 (4), 100019. <https://doi.org/10.1016/j.aosl.2020.100019>.
- 16 Zhang, L.; Huang, J.; Yu, H.; **Liu, X.**; Wei, Y.; Lian, X.; Liu, C.; Jing, Z. Optimal Parameterization of COVID-19 Epidemic Models. *Atmospheric and Oceanic Science Letters* 2021, 14 (4), 100024. <https://doi.org/10.1016/j.aosl.2020.100024>.
- 15 Huang, J.; **Liu, X.**; Zhang, L.; Zhao, Y.; Wang, D.; Gao, J.; Lian, X.; Liu, C. The Oscillation-Outbreaks Characteristic of the COVID-19 Pandemic. *Natl. Sci. Rev.* 2021, 8 (8), nwab100. <https://doi.org/10.1093/nsr/nwab100>.
- 14 Li, C.; Huang, J.; Ding, L.; **Liu, X.**; Han, D.; Huang, J. Estimation of Oceanic and Land Carbon Sinks Based on the Most Recent Oxygen Budget. *Earth's Future* 2021, 9 (7). <https://doi.org/10.1029/2021EF002124>.
- 13 Wei, Y.; Wu, J.; Huang, J.; **Liu, X.**; Han, D.; An, L.; Yu, H.; Huang, J. Declining Oxygen Level as an Emerging Concern to Global Cities. *Environ. Sci. Technol.* 2021, 55 (12), 7808–7817. <https://doi.org/10.1021/acs.est.1c00553>. **Journal Cover**
- 12 Huang, J.; **Liu, X.**; He, Y.; Shen, S.; Hou, Z.; Li, S.; Li, C.; Yao, L.; Huang, J. The Oxygen Cycle and a Habitable Earth. *Sci. China Earth Sci.* 2021, 64 (4), 511–528. <https://doi.org/10.1007/s11430-020-9747-1>. **Journal Cover**
- 11 Lian, X.; Huang, J.; Zhang, L.; Liu, C.; **Liu, X.**; Wang, L. Environmental Indicator for COVID-19 Non-Pharmaceutical Interventions. *Geophys. Res. Lett.* 2021, 48 (2). <https://doi.org/10.1029/2020GL090344>.
- 10 Yang, F.; Huang, J.; He, Q.; Zheng, X.; Zhou, C.; Pan, H.; Huo, W.; Yu, H.; **Liu, X.**; Meng, L.; Han, D.; Ali, M.; Yang, X. Impact of Differences in Soil Temperature on the Desert Carbon Sink. *Geoderma* 2020, 379, 114636. <https://doi.org/10.1016/j.geoderma.2020.114636>.
- 9 Huang, J.; Zhang, L.; **Liu, X.**; Wei, Y.; Liu, C.; Lian, X.; Huang, Z.; Chou, J.; Liu, X.; Li, X.; Yang, K.; Wang, J.; Liang, H.; Gu, Q.; Du, P.; Zhang, T. Global Prediction System for COVID-19 Pandemic. *Science Bulletin* 2020, 65 (22), 1884–1887. <https://doi.org/10.1016/j.scib.2020.08.002>.
- 8 Huang, J.; Yu, H.; Han, D.; Zhang, G.; Wei, Y.; Huang, J.; An, L.; **Liu, X.**; Ren, Y. Declines in Global Ecological Security under Climate Change. *Ecol. Indic.* 2020, 117 (April), 106651. <https://doi.org/10.1016/j.ecolind.2020.106651>.
- 7 Li, C.; Huang, J.; Ding, L.; **Liu, X.**; Yu, H.; Huang, J. Increasing Escape of Oxygen from Oceans Under Climate Change. *Geophys. Res. Lett.* 2020, 47 (11), 1–8. <https://doi.org/10.1029/2019GL086345>.
- 6 Yang, F.; Huang, J.; Zhou, C.; Yang, X.; Ali, M.; Li, C.; Pan, H.; Huo, W.; Yu, H.; **Liu, X.**; Zheng, X.; Han, D.; He, Q.; Meng, L.; Chang, J. Taklimakan Desert Carbon-Sink Decreases under Climate Change. *Science Bulletin* 2020, 65 (6), 431–433. <https://doi.org/10.1016/j.scib.2019.12.022>.
- 5 Bi, L.; He, Y.; Huang, J.; Li, Y.; Guan, X.; **Liu, X.** The Global Response of Temperature to High-Latitude Vegetation Greening in a Two-Dimensional Energy Balance Model. *Atmospheric and Oceanic Science Letters* 2020, 13 (1), 80–87. <https://doi.org/10.1080/16742834.2020.1696650>.

- 4 Yu, H.; Wei, Y.; Zhang, Q.; **Liu, X.**; Huang, J.; Feng, T.; Zhang, M. Multi-model Assessment of Global Temperature Variability on Different Time Scales. *Int J Climatol* 2020, 40 (1), 273–291. <https://doi.org/10.1002/joc.6209>.
- 3 Zhang, M.; Yu, H.; Huang, J.; Wei, Y.; **Liu, X.**; Zhang, T. Comparison of Extreme Temperature Response to 0.5 °C Additional Warming between Dry and Humid Regions over East–Central Asia. *Int J Climatol* 2019, 39 (7), 3348–3364. <https://doi.org/10.1002/joc.6025>.
- 2 Huang, J.; Huang, J.; **Liu, X.**; Li, C.; Ding, L.; Yu, H. The Global Oxygen Budget and Its Future Projection. *Science Bulletin* 2018, 63 (18), 1180–1186. <https://doi.org/10.1016/j.scib.2018.07.023>.
- 1 Wei, Y.; Yu, H.; Huang, J.; He, Y.; Yang, B.; Guan, X.; **Liu, X.** Comparison of the Pacific Decadal Oscillation in Climate Model Simulations and Observations. *Int. J. Climatol* 2018, 38, e99–e118. <https://doi.org/10.1002/joc.5355>.